Name:Shruti Santosh Pol

Roll no: 30

**CODE**

```
import numpy as np

# Create an array
a = np.array([1, 4, 9, 16, 25])
b = np.array([2, 3, 4, 5, 6])

print("Array A:", a)
print("Array B:", b)

print("Arithmetic functions")
# 1. Addition
print("Addition:", np.add(a, b))

# 2. Subtraction
print("Subtraction:", np.subtract(a, b))

# 3. Multiplication
print("Multiplication:", np.multiply(a, b))

# 4. Division
print("Division:", np.divide(a, b))
```

```
# 5. Floor Division
print("Floor Division:", np.floor_divide(a, b))

# 6. Modulus
print("Modulus:", np.mod(a, b))

# 7. Power
print("Power:", np.power(a, 2))
print('\n')

print("Mathematical Functions")
# 8. Square Root
print("Square Root:", np.sqrt(a))

# 9. Absolute Value
print("Absolute Value:", np.abs(-a))

# 10. Exponential
print("Exponential:", np.exp(a))

# 11. Natural Logarithm
print("Natural Log:", np.log(a))

# 12. Log Base 10
print("Log Base 10:", np.log10(a))
print("\n")
```

```
print("Trigonometric Functions")
# 13. Sine
print("Sine:", np.sin(a))

# 14. Cosine
print("Cosine:", np.cos(a))

# 15. Tangent
print("Tangent:", np.tan(a))
print("\n")

print("Comparison Functions")
# 16. Maximum
print("Maximum:", np.maximum(a, b))

# 17. Minimum
print("Minimum:", np.minimum(a, b))

# 18. Greater Than
print("Greater Than:", np.greater(a, b))

# 19. Less Than
print("Less Than:", np.less(a, b))

# 20. Equal
```

print("Equal:", np.equal(a, b))

**OUTPUT**

```
print("Less Than:", np.less(a, b))

# 20. Equal
print("Equal:", np.equal(a, b))
```

```
Array A: [ 1  4  9 16 25]
Array B: [2 3 4 5 6]
Arithmetic functions
Addition: [ 3  7 13 21 31]
Subtraction: [-1  1  5 11 19]
Multiplication: [  2  12  36  80 150]
Division: [0.5        1.33333333 2.25       3.2        4.16666667]
Floor Division: [0 1 2 3 4]
Modulus: [1 1 1 1 1]
Power: [  1  16  81 256 625]


Mathematical Functions
Square Root: [1. 2. 3. 4. 5.]
Absolute Value: [ 1  4  9 16 25]
Exponential: [2.71828183e+00 5.45981500e+01 8.10308393e+03 8.88611052e+06
 7.20048993e+10]
Natural Log: [0.         1.38629436 2.19722458 2.77258872 3.21887582]
Log Base 10: [0.         0.60205999 0.95424251 1.20411998 1.39794001]


Trigonometric Functions
Sine: [ 0.84147098 -0.7568025   0.41211849 -0.28790332 -0.13235175]
Cosine: [ 0.54030231 -0.65364362 -0.91113026 -0.95765948  0.99120281]
Tangent: [ 1.55740772  1.15782128 -0.45231566  0.30063224 -0.13352641]


Comparison Functions
Maximum: [ 2  4  9 16 25]
Minimum: [1 3 4 5 6]
Greater Than: [False  True  True  True  True]
Less Than: [ True False False False False]
Equal: [False False False False False]
```